

The Finitude Theorem

Why Infinity Cannot Be Physical

William J. Steinmetz III

Framework: Foundational Ternary Dynamics v5.28

Abstract

We prove that no physical observable can be infinite within a framework where space is a discrete lattice $\mathbb{Z}^3$, time advances in integer ticks $t \in \mathbb{N}$, states belong to a finite alphabet $\{-1, 0, +1\}$, and dynamics are local (26-neighbor Moore neighborhood). The *Finitude Theorem* establishes that for any lattice configuration reachable from finite initial data by local deterministic rules, every computable quantity is bounded by an explicit function of the initial energy E_0 and elapsed time t . No singularity, divergence, or blow-up can occur — not because we lack the mathematics to control them, but because the substrate cannot represent them. We distinguish the *ontic* domain (what physically exists: always finite) from the *epistemic* domain (what we can reason about: includes infinity as a limit concept). Infinity is a property of our mathematical language, not of the reality it describes. Every “infinity” that appears in physics — UV divergences, black hole singularities, Navier–Stokes blow-up, the cosmological singularity — is traced to the use of $\mathbb{R}^3$ where $\mathbb{Z}^3$ is the actual substrate. The continuum is an extraordinarily good approximation, but the approximation *itself* is what creates the pathologies.

Contents

1	The Ontic–Epistemic Distinction	2
2	The Finitude Theorem	2
3	Why Infinity Cannot Be Constructed	3
4	The Catalog of Infinities: Where They Come From	4
4.1	UV divergences in quantum field theory	4
4.2	Black hole singularities	4
4.3	Navier–Stokes blow-up	5
4.4	The cosmological singularity	5
4.5	Hilbert space dimension	5
5	The Epistemic Role of Infinity	5
6	Formal Structure: The Ontic–Epistemic Partition	6
7	Implications for the Millennium Problems	6
7.1	Navier–Stokes existence and smoothness	6
7.2	Yang–Mills existence and mass gap	7
7.3	Riemann hypothesis	7
8	Objections and Responses	7
9	Conclusion	8

1. The Ontic–Epistemic Distinction

Definition 1.1 (Ontic domain). The *ontic domain* is the domain of what physically exists: the actual configuration of the lattice at each tick. Every element of the ontic domain is a triple $(\mathbf{v}, t, q)$ where $\mathbf{v} \in \mathbb{Z}^3$ is a voxel address, $t \in \mathbb{N}$ is a tick index, and $q = (s, \mathbf{J}, \mathbf{w})$ is the local state (ternary charge, flux vector, velocity).

Definition 1.2 (Epistemic domain). The *epistemic domain* is the domain of mathematical reasoning about the ontic domain. It includes all of classical and modern mathematics: the real numbers $\mathbb{R}$, the complex numbers, infinite-dimensional Hilbert spaces, distributions, limits, and the concept of infinity itself.

Selection 1.3 (The asymmetry). The ontic domain is always finite. The epistemic domain is not. The real line $\mathbb{R}$ is an epistemic object. It contains irrational numbers $(\sqrt{2}, \pi, e)$ that cannot be written as finite strings. It contains uncountably many points between any two rational numbers. No physical measurement can distinguish π from a sufficiently close rational approximation.

The lattice $\mathbb{Z}^3$ is an ontic object. Every voxel has a finite integer address. Every state is one of three values. Every flux vector has components that are computed by finite arithmetic (addition, multiplication, division of finite-precision numbers).

Remark 1.4. This is not a philosophical preference. It is a structural consequence of the five postulates. If space is $\mathbb{Z}^3$, then the concept “a point at position $(\sqrt{2}, \pi, e)$ ” has no ontic referent — there is no such voxel. The address $(\sqrt{2}, \pi, e)$ exists in the epistemic domain (we can write it, reason about it, compute with it) but not in the ontic domain (no voxel is located there).

2. The Finitude Theorem

Theorem 2.1 (Finitude Theorem — Main Result). *[THEOREM] Let $\Lambda \subset \mathbb{Z}^3$ be a finite lattice with $|\Lambda| = N$ voxels. Let the initial configuration satisfy $E_0 := \sum_{\mathbf{v} \in \Lambda} \frac{1}{2}(|\mathbf{J}(\mathbf{v}, 0)|^2 + |\mathbf{w}(\mathbf{v}, 0)|^2) < \infty$. Then for every tick $t \in \mathbb{N}$ and every voxel $\mathbf{v} \in \Lambda$:*

1. $|s(\mathbf{v}, t)| \leq 1$,
2. $|\mathbf{J}(\mathbf{v}, t)| \leq B_J(E_0, N, t) < \infty$,
3. $|\mathbf{w}(\mathbf{v}, t)| \leq B_w(E_0, N, t) < \infty$,
4. *Every derived quantity (energy, vorticity, pressure, curvature, force) is bounded by an explicit function of E_0 , N , and t .*

No physical observable can be infinite.

Proof. The proof proceeds by induction on t .

Base case ($t = 0$). By hypothesis, $E_0 < \infty$, so $|\mathbf{J}(\mathbf{v}, 0)| \leq \sqrt{2E_0}$ and $|\mathbf{w}(\mathbf{v}, 0)| \leq \sqrt{2E_0}$ for all $\mathbf{v}$. Statement (1) holds by the ternary axiom ($s \in \{-1, 0, +1\}$).

Inductive step. Assume all quantities are bounded at tick t . The update proceeds through six phases:

Phase 1 (read): $\Delta \mathbf{J}(\mathbf{v}) = C^2 \nabla_L^2 \mathbf{J}(\mathbf{v}) + g_c \nabla_L s(\mathbf{v})$. The lattice Laplacian averages over 6 neighbors:

$$|\nabla_L^2 \mathbf{J}(\mathbf{v})| \leq \sum_{\mu=1}^3 |J(\mathbf{v} + \hat{e}_\mu) + J(\mathbf{v} - \hat{e}_\mu) - 2J(\mathbf{v})| \leq 6 \cdot 2 \max_{\mathbf{v}'} |\mathbf{J}(\mathbf{v}', t)|.$$

Hence $|\Delta \mathbf{J}(\mathbf{v})| \leq 12C^2 \cdot B_J(t) + 2g_c := \delta(t) < \infty$.

Phase 2 (write): $\mathbf{w} += \Delta \mathbf{J}$ and $\mathbf{J} += \mathbf{w}$. Hence $|\mathbf{w}(\mathbf{v}, t+1)| \leq B_w(t) + \delta(t)$ and $|\mathbf{J}(\mathbf{v}, t+1)| \leq B_J(t) + B_w(t) + \delta(t)$. Both are finite.

Phase 3 (Gauss project): The Poisson solve computes ϕ with $\nabla_L^2 \phi = \nabla_L \mathbf{J} - s$. Since $|\nabla_L \mathbf{J}| \leq 6 \max |\mathbf{J}|$ and $|s| \leq 1$, the source is bounded. On a finite lattice, the Poisson equation with bounded source has a unique bounded solution. The correction $\mathbf{J} -= \nabla_L \phi$ does not increase $\max |\mathbf{J}|$ (it projects onto the divergence-free subspace).

Phases 4–5 (forces, movement): Forces are polynomials in $\mathbf{J}$, s , and lattice gradients of bounded fields. Velocities are bounded by $C = 1/\sqrt{3}$. All updates produce finite values from finite inputs.

Phase 6 (tick++): $t += 1$, a finite operation.

Therefore the bounds $B_J(t+1)$ and $B_w(t+1)$ are finite, completing the induction.

For statement (4): every derived quantity — energy $\frac{1}{2}|\mathbf{J}|^2$, vorticity $|\nabla_L \times \mathbf{J}|$, pressure $|\nabla_L^2 \phi|$, force $|\mathbf{F}|$ — is a bounded function of $\mathbf{J}$, $\mathbf{w}$, s , and their lattice gradients, all of which are bounded. $\square$

Remark 2.2 (The bound is explicit). We can write $B_J(E_0, N, t)$ explicitly. In the worst case (no dissipation, maximum amplification at every tick):

$$B_J(t) \leq (1 + 12C^2)^t \cdot \sqrt{2E_0} + \frac{2g_c}{12C^2} [(1 + 12C^2)^t - 1].$$

With $C^2 = 1/3$, the growth factor is $(1 + 4)^t = 5^t$ per tick. This is exponential in t — fast, but *finite* for every finite t . In practice, the CFL dissipation prevents exponential growth; the bound is conservative.

3. Why Infinity Cannot Be Constructed

The Finitude Theorem is not merely a bound. It is a *structural impossibility result*: the lattice cannot represent infinity.

Proposition 3.1 (No infinite address). *[THEOREM] There is no voxel at position ∞ . Every voxel address $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{Z}^3$ has finite integer coordinates.*

Proposition 3.2 (No infinite state). *[THEOREM] The state alphabet $\{-1, 0, +1\}$ does not contain ∞ . No voxel can be in state “infinity.”*

Proposition 3.3 (No infinite velocity). *[THEOREM] The CFL bound $|\mathbf{v}_{\text{vel}}| \leq 1/\sqrt{3}$ prevents infinite speed. A signal would need to reach infinitely many voxels in finite time to propagate infinitely fast, but the Moore neighborhood contains exactly 26 neighbors — a finite set.*

Proposition 3.4 (No infinite energy density). *[THEOREM] The energy at a single voxel $\frac{1}{2}|\mathbf{J}(\mathbf{v}, t)|^2$*

is bounded by $B_J(t)^2/2 < \infty$ for all finite t . Infinite energy density would require $|\mathbf{J}| \rightarrow \infty$, which the Finitude Theorem excludes.

Proposition 3.5 (No infinite information). *[THEOREM] In the causal past of an observer at tick t , the number of voxels is $N_{\text{causal}}(t) \leq \frac{4\pi}{3}(Ct)^3 = \frac{4\pi}{3 \cdot 3\sqrt{3}}t^3 < \infty$. The total information content is $N_{\text{causal}} \times (\log_2 3 + 3 \times b_J)$ bits, where b_J is the precision of each flux component. This is finite for finite t at any fixed precision.*

Theorem 3.6 (Impossibility of physical infinity). *[THEOREM given the five postulates] Under the five postulates of FTD (discrete space, discrete time, ternary states, local causality, determinism), no physically realizable configuration can contain, produce, or represent an infinite quantity.*

More precisely: let $Q : \Lambda \times \mathbb{N} \rightarrow \mathbb{R}$ be any computable function of the lattice configuration. If Q depends only on the state and flux values within a finite causal region, then $|Q(\mathbf{v}, t)| < \infty$ for all $\mathbf{v} \in \Lambda$ and $t \in \mathbb{N}$.

Proof. Every computable function of a finite set of finite inputs produces a finite output. The lattice configuration within any finite causal region consists of finitely many voxels, each carrying a finite state and finite flux values (Finitude Theorem). Therefore Q is finite. $\square$

4. The Catalog of Infinities: Where They Come From

Every infinity in physics arises from the same source: the use of $\mathbb{R}^n$ where $\mathbb{Z}^n$ is the actual substrate. We catalog the major instances.

4.1 UV divergences in quantum field theory

Continuum pathology	Cause	Lattice resolution
Electron self-energy $\int \frac{d^4 k}{k^2} \rightarrow \infty$	Integration over $\mathbb{R}^4$ to $ k \rightarrow \infty$	Brillouin zone $[-\pi, \pi]^3$ is compact; integral finite
Vacuum energy $\sum_k \frac{1}{2} \hbar \omega_k \rightarrow \infty$	Infinite modes on $\mathbb{R}^3$	Finite modes on $\mathbb{Z}^3$ (N voxels $\rightarrow N$ modes)
Landau pole $\alpha(\mu) \rightarrow \infty$	Running coupling hits pole at finite μ	Lattice spacing provides natural UV cutoff; pole is unphysical

4.2 Black hole singularities

In general relativity on $\mathbb{R}^{3,1}$, the Schwarzschild metric has $g_{tt} \rightarrow 0$ and curvature $R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \rightarrow \infty$ as $r \rightarrow 0$. On the lattice, “ $r = 0$ ” is a single voxel. The curvature at that voxel is a finite function of the flux field in its neighborhood (26 finite neighbors with finite fields). The curvature divergence is an artifact of taking $r \rightarrow 0$ in a continuous metric — a limit that has no ontic referent.

4.3 Navier–Stokes blow-up

The continuum Navier–Stokes equations on $\mathbb{R}^3$ may develop singularities because the gradient $\nabla \mathbf{u}$ is not controlled by $\|\mathbf{u}\|_{L^\infty}$. On the lattice, $|\nabla_L \mathbf{J}| \leq 2 \max |\mathbf{J}|$ — the gradient is always bounded by the pointwise values. Blow-up requires infinite energy density at a point; the Finitude Theorem excludes this.

4.4 The cosmological singularity

The Big Bang singularity in FRW cosmology has $a(t) \rightarrow 0$ and $\rho \rightarrow \infty$ as $t \rightarrow 0$. On the lattice, “ $t = 0$ ” is the first tick. The initial configuration has finite energy $E_0 < \infty$. The density $\rho = E_0/N$ is finite for any nonzero number of voxels. The singularity is an artifact of extrapolating continuous FRW backwards through scales below the lattice spacing.

4.5 Hilbert space dimension

Standard quantum mechanics uses infinite-dimensional Hilbert spaces $\mathcal{H} = L^2(\mathbb{R}^3)$. On the lattice, the state space of N voxels with ternary states is 3^N -dimensional (finite). The flux field adds continuous degrees of freedom, but at each voxel the flux is a finite-precision vector. The effective Hilbert space dimension is large but finite.

5. The Epistemic Role of Infinity

If infinity is not physical, why is it so useful in mathematics and physics?

Selection 5.1 (Infinity as a limit concept). Infinity is a property of *sequences*, not of *values*. The statement “ $\lim_{n \rightarrow \infty} (1 + 1/n)^n = e$ ” does not assert that the sequence reaches e at $n = \infty$. It asserts that for any desired precision $\epsilon > 0$, there exists a finite N such that $|(1 + 1/n)^n - e| < \epsilon$ for all $n > N$. Every application of infinity in physics can be rephrased as a statement about finite approximations.

Selection 5.2 (Infinity as computational convenience). Integrals over $\mathbb{R}^3$ are easier to compute than sums over $\mathbb{Z}^3$. The continuum provides Fourier transforms, distribution theory, and differential geometry — powerful tools that would be cumbersome to replicate in discrete mathematics. Using $\mathbb{R}^3$ as an approximation to $\mathbb{Z}^3$ is like using floating-point arithmetic to approximate integer arithmetic: it is faster and more elegant, at the cost of introducing artifacts (overflow, underflow, rounding errors) that do not exist in the exact integer computation.

In physics, the “rounding errors” of the continuum approximation are:

- UV divergences (the analogue of overflow),
- IR divergences (the analogue of underflow),
- Singularities (the analogue of division by zero).

None of these exist in the discrete computation.

Remark 5.3 (The unreasonable effectiveness of the continuum). The continuum approximation works spectacularly well at scales $\gg a$ (the lattice spacing). At human scales, $(a/L)^2 \sim 10^{-70}$, so the continuum is accurate to extraordinary precision. The pathologies (divergences, singularities) only appear at scales $\sim a$, where the approximation breaks down. This is analogous to Newtonian gravity: it works perfectly at human scales and only fails at relativistic speeds or strong fields. The failure is a property of the approximation, not of reality.

6. Formal Structure: The Ontic–Epistemic Partition

Definition 6.1 (Ontic–epistemic partition). Every mathematical statement about physics can be classified as:

Domain	Character	Infinity status
Ontic	What physically exists; lattice configurations; measurable quantities	Always finite
Epistemic	Mathematical models; limit processes; formal reasoning	May include infinity

Theorem 6.2 (Ontic closure under finite operations). *[THEOREM] The ontic domain is closed under the FTD update rules. Starting from a finite configuration, every update produces a finite configuration. No finite sequence of local operations can produce an infinite value.*

Proof. The update rules (phases 1–6) are compositions of addition, subtraction, multiplication, division (with bounded denominators), and comparison. Each operation maps finite inputs to finite outputs. By composition and induction, the result is finite. $\square$

Corollary 6.3 (Infinity is strictly epistemic). *Every occurrence of ∞ in a physical theory is a feature of the epistemic model, not of the ontic reality. Removing ∞ from a physical theory (by replacing $\mathbb{R}^3$ with $\mathbb{Z}^3$ and $\mathbb{R}$ with $\mathbb{N}$) eliminates all singularities and divergences without loss of predictive accuracy at scales $\gg a$.*

7. Implications for the Millennium Problems

The Finitude Theorem has immediate consequences for three of the Clay Millennium Prize Problems:

7.1 Navier–Stokes existence and smoothness

Continuum question: Can solutions blow up in finite time?

Finitude answer: No. Blow-up means $\|\mathbf{u}\|_{L^\infty} \rightarrow \infty$, which requires infinite energy density at a point. The Finitude Theorem (Proposition 3.4) excludes this. The question is an artifact of using $\mathbb{R}^3$ where $\mathbb{Z}^3$ is the substrate.

7.2 Yang–Mills existence and mass gap

Continuum question: Does a non-trivial quantum Yang–Mills theory exist on $\mathbb{R}^4$ with mass gap $\Delta > 0$?

Finitude answer: The theory exists on $\mathbb{Z}^3 \times \mathbb{N}$ with $\Delta = K_B = 0.511$ MeV. UV finiteness is automatic (compact Brillouin zone). IR finiteness follows from the mass gap. The question of “existence on $\mathbb{R}^4$ ” is asking whether the epistemic model ($\mathbb{R}^4$) can support a structure that the ontic substrate ($\mathbb{Z}^3 \times \mathbb{N}$) already provides.

7.3 Riemann hypothesis

Epistemic question: Do all non-trivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = 1/2$?

The Riemann zeta function is a purely epistemic object — an infinite series $\sum n^{-s}$. However, the physical instantiation of the critical structure (the theta function identity $G^* = \sqrt{2\pi} \cdot \theta_3(e^{-\pi})^2$) is finite: the nome $|q| = e^{-\pi} \approx 0.043$ ensures geometric convergence. The infinite series $\theta_3 = 1 + 2 \sum_{n=1}^{\infty} q^{n^2}$ converges to a finite value. The “infinity” in the series is an epistemic artifact of writing a convergent sum as a limit.

8. Objections and Responses

Objection 1: But mathematics uses infinity everywhere

Response: Yes, and it should continue to do so. The Finitude Theorem does not restrict mathematical practice. It restricts the *interpretation* of mathematical objects: infinity is a tool for reasoning about finite quantities (via limits, completeness, compactness), not a claim about physical existence. We use π freely in calculations, knowing that the lattice computes with finite-precision approximations.

Objection 2: The universe might be infinite in spatial extent

Response: Even if $\mathbb{Z}^3$ is infinite (as the postulate states), no observer can access an infinite number of voxels. The causal past $\mathcal{P}_i(t)$ contains at most $(Ct)^3$ voxels at tick t . An observer at tick $t = 10^{60}$ (roughly the current age in Planck ticks) has access to $\sim 10^{180}$ voxels — enormous but finite. The lattice may be infinite in extent, but every observable quantity is computed from a finite causal region.

Objection 3: Quantum mechanics requires infinite-dimensional Hilbert spaces

Response: On the lattice, the Hilbert space for N ternary voxels is 3^N -dimensional. At $N = 10^{180}$, the dimension $3^{10^{180}}$ is unimaginably large but finite. The infinite-dimensional $L^2(\mathbb{R}^3)$ is the continuum approximation, useful for calculation but not physically necessary. Every quantum mechanical prediction (expectation values, transition amplitudes, cross sections)

can be computed in the finite-dimensional lattice Hilbert space.

Objection 4: This is just ultraviolet regularization with extra philosophy

Response: No. UV regularization introduces a cutoff into a continuum theory and then takes the cutoff to infinity (the renormalization group flow). FTD does not take any limit. The lattice spacing is fundamental and fixed. There is no cutoff to remove, no limit to take, no renormalization needed. The difference is ontological, not technical: the lattice is not a regularization of a continuum theory; the continuum is an approximation to a lattice theory.

9. Conclusion

The Finitude Theorem establishes a clean partition of mathematical physics into two domains:

	Ontic (physical)	Epistemic (mathematical)
Substrate	$\mathbb{Z}^3 \times \mathbb{N}$	$\mathbb{R}^{3,1}$
States	$\{-1, 0, +1\}$	(infinite-dimensional Hilbert space)
Quantities	Always finite	May be infinite (limits, series, integrals)
Singularities	Impossible	Artifacts of continuum idealization
Divergences	Impossible	Artifacts of integration over $\mathbb{R}^n$

No experiment in the history of science has measured an infinite quantity. No physical system has exhibited a singularity — only mathematical models of physical systems have. The Finitude Theorem explains why: the physical substrate is discrete, local, and bounded. Infinity is a property of our mathematical language, not of the reality it describes.

The implications are concrete:

- **Navier–Stokes:** Blow-up cannot occur on $\mathbb{Z}^3$. The question dissolves.
- **Yang–Mills:** The mass gap is automatic ($K_B > 0$). UV divergences don’t arise.
- **Black holes:** The singularity at $r = 0$ is a single voxel with finite curvature.
- **The Big Bang:** The first tick has finite energy. No initial singularity.
- **QFT renormalization:** Not needed. The lattice is the theory, not a regularization.

Infinity has served physics magnificently as an epistemic tool for three centuries. The Finitude Theorem does not banish it from mathematics. It simply recognizes that infinity lives in the map, not in the territory.

References

- [1] W. J. Steinmetz III, *Foundational Ternary Dynamics: The Complete Specification*, v5.28, 2026.

- [2] D. Hilbert, Über das Unendliche, *Mathematische Annalen* **95** (1926), 161–190.
- [3] D. Zeilberger, “Real” analysis is a degenerate case of discrete analysis, *Proceedings of the Sixth International Conference on Difference Equations* (2004), 25–34.
- [4] E. Fredkin, Finite Nature, *Proceedings of the XXVIIth Rencontre de Moriond* (1992).
- [5] G. 't Hooft, The cellular automaton interpretation of quantum mechanics, *Fundamental Theories of Physics* **185** (2016).
- [6] S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002.
- [7] R. Penrose, *The Road to Reality*, Vintage Books, 2004.